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Rethinking news framing with large language models

Amir Tohidi[1]✉, Samar Haider[1] & Duncan J. Watts[1,2,3]

Mainstream media, with its broad reach, plays a central role in shaping public opinion and thus warrants close scrutiny. Subtle forms of media bias–such as selective fact presentation and tone–can meaningfully influence public attitudes, even when reporting remains factually accurate. Although effects such as these have been widely studied by scholars of framing, much of the existing research focuses on specific topics and relies on manually constructed or pre-existing frames, limiting both scalability and generalizability. Here we introduce a novel framework that leverages large language models (LLMs) to generate synthetic news articles by systematically varying the selection and tone of the content while holding factual accuracy and other features constant. We evaluate the impact of these alternative framings in a large, pre-registered randomized experiment (N = 2,141), and find that selective presentation of accurate information can significantly shift individuals' policy views and emotional responses across a diverse collection of topics. These effects are consistently stronger for negative than positive framings and are more pronounced among individuals who say they are less informed about the topic. Our findings demonstrate the persuasive power of subtle bias in mainstream news as well as the value of LLMs as tools for scalable, controlled investigations of media effects.

Keywords Media bias, News framing, Persuasion, LLMs, Experiment

In the era of social media and digital information platforms, traditional news outlets remain the dominant source of news for most people[1] and continue to shape political preferences and attitudes[2]. Although such news organizations are less prone to the kind of misinformation that is widely studied in the context of social media[3–6], they can still suffer from multiple forms of bias, such as selective emphasis on certain stories, events, facts, and perspectives, as well as bias that is introduced via editorial slant and tone[7].

A long tradition of work on what political scientists and communications scholars refer to as "framing"[8–13] has demonstrated the potential impact of the biased presentation of factually accurate content. However, traditional approaches have been impeded by two important methodological limitations. First, experimental designs have generally relied on manually crafted stimuli, making it difficult to independently vary the many potential contributing factors such such as content, tone, or writing style, while keeping all else constant. Second, prior work has tended to focus on isolated contexts—for example free speech[14], immigration[15–17], gun policy[18], civil protests[19,20], or political preferences[21]–which can vary on multiple dimensions, thereby limiting the generalizability of findings.

In this paper we address both of these limitations with a novel framework that leverages the capabilities of large language models (LLMs) to study the effect of biased news articles across a broad range of contemporary issues. By utilizing vast training datasets, LLMs can discern subtle variations in tone, emphasis, and narrative structure[22–24]. As a result, they have demonstrated human-level ability to identify nuanced semantic relationships[25–27] and subtle frames in the text of news articles[28,29]; however, these applications have largely been limited to observational analysis and frame detection, rather than experimental testing. LLMs have also been shown to be effective in persuading human interlocutors by communicating factual information[30–34]; however, these studies have focused on correcting false beliefs and conspiratorial thinking rather than investigating the effects of biased presentation of facts.

Our LLM framework comprises multiple stages. First, we compiled a collection of real-world news articles published by 10 reputable publishers with varying partisan orientations on a set of seven widely covered events that were manually selected to cover a diverse range of topics, incorporating politics, economics, and culture, as well as differing levels of public awareness (see ref.[35] for details on collection of original news articles and SI 1.2 for full list of events). Second, we used GPT-4o to classify every sentence of each article on two dimensions:

[1]Department of Computer and Information Science, University of Pennsylvania, Philadelphia, PA 19104, USA. [2]Annenberg School of Communication, University of Pennsylvania, Philadelphia, PA 19104, USA. [3]Operations, Information, and Decisions Department, University of Pennsylvania, Philadelphia, PA 19104, USA. ✉email: atohidi@seas.upenn.edu

type, defined as fact, quote, or opinion; and tone, defined as positive, neutral, or negative. Here we emphasize that our definition of "fact" is a sentence that has the form of a factual statement, not that it is a true fact[36]. We did not conduct any independent fact-checking on the statements, but note that all statements were sourced from mainstream media outlets, which generally do employ fact checking. Consequently, the resulting stimuli (described next) should be viewed as exhibiting a level of factual accuracy comparable to that of typical articles published by mainstream media organizations. We also used GPT-4o to remove duplicates (or near duplicates), generating a collection of unique sentences, and to rank them in terms of informativeness and relevance to the event in question. Third, we then used these curated sets of sentences as input for GPT-4o model, which we tasked with crafting three versions of each news story that varied specifically on the dimension of tone: one positive, one negative, and one neutral. We note that we did not ask GPT to rewrite the sentences to change their tone; in fact, by setting the model temperature to zero, we kept the sentences as close as possible to their original form. Instead tone was used as a selection criterion to determine which factual, quoted, or opinion statements are included in each condition. Consequently, the positive, neutral, and negative versions differ not only in tone but also in the specific information presented, reflecting how journalists might introduce bias into an article by selecting facts and quotes that support the narrative they wish to advance. Although we believe this approach increases external validity, it also means that our estimated effects are a combination of both variation in information selection and tone variation.

To maintain consistency of formatting across stimuli, we instructed GPT to draw exclusively from the subset of sentences labeled with the assigned tone and to combine them in the same ratio of facts to quotes to opinions. As just noted, the generated articles reproduce the original sentences from mainstream news sources almost verbatim by setting the model temperature parameter to zero, and also we manually reviewed them to ensure fidelity to the source text. To minimize other sources of variance such as writing style and quality, all articles were created with the same model and hyperparameters[37]. In this way, our approach yielded three versions of each story that were almost similar in length, composition, writing style, and factual accuracy, yet presented the event in question in very different valences. Figure 1 provides a visual summary of stages two and three–information extraction and article generation. For further implementation details, see the "Methods" section.

Armed with these synthetic articles we then conducted a large-scale randomized between-subjects experiment (N = 2,141, see "Methods" for more details) in which participants were randomly assigned to one of

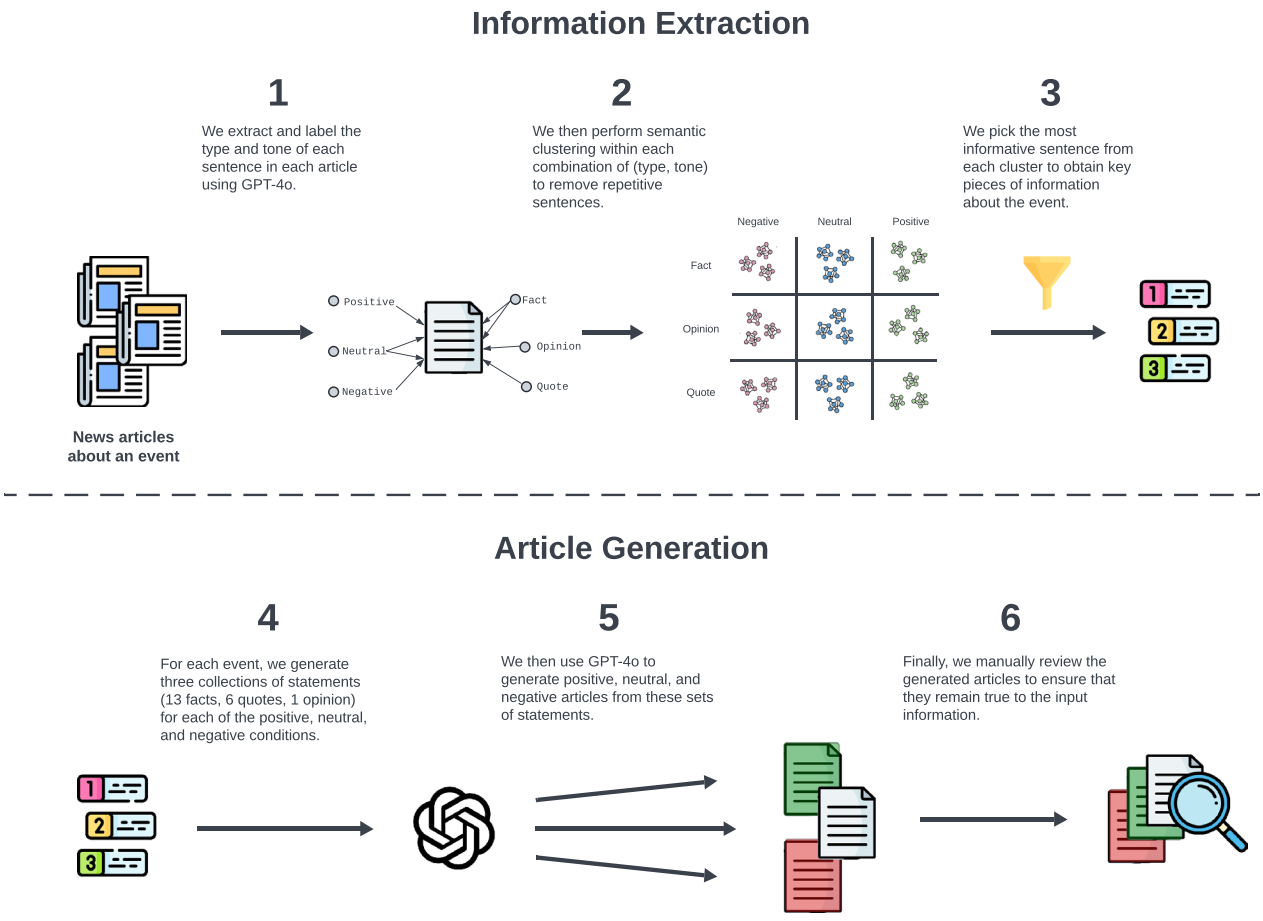


Fig. 1. Stimuli generation pipeline. This diagram outlines the step-by-step process of generating synthetic articles from news content. This pipeline leverages advances in LLMs to create scalable and controlled stimuli for assessing framing effects across a diverse range of topics.

three experimental conditions (*Positive*, *Neutral*, *Negative*). To prevent potential spillover effects across events, each participant was assigned to only one of the selected events. Before exposure to any articles, we collected demographic information (age, race, gender, partisanship) and data on respondents' general media engagement. To reduce demand effects, participants were also presented with instructions indicating that the article they would read presented only one perspective and that other perspectives might be equally consistent with the facts. After reading the article, participants completed a feeling thermometer question which asked them to rate their "feelings toward the main subject of the article", and were also asked about their prior informedness about the subject, and to evaluate the article's tone and informativeness. In addition, they answered a series of six event-specific questions designed to gauge their views on various aspects related to the event. All responses were recorded on a six-point Likert scale. The full list of questions and the synthetic articles can be found in the project's OSF page (OSF link: https://osf.io/9g7sq/?view_only=189a8b9b2f644433bdbfcd4fc5c63ffe).

Our findings demonstrate that even when factual accuracy is preserved, selectively emphasizing negatively toned information can meaningfully shift public attitudes and emotional responses. This effect, when considered alongside the well-documented negativity bias in mainstream media, raises important concerns about how news consumption shapes perceptions of the world. Methodologically, our study introduces a scalable and controlled framework for testing framing effects using large language models, enabling a level of experimental precision and topical breadth that was previously infeasible. In addition to advancing theoretical understanding of media influence, this approach invites further inquiry into media literacy and potential policy responses aimed at reducing the impact of subtle framing biases and fostering a more critically engaged public.

Before proceeding, we note that our focus on varying information selection based on its tone is most closely aligned with what is called emphasis framing[38,39], which refers to "selecting some aspects of a perceived reality and making them more salient"[9]. Specifically, our manipulation involved systematically selecting subsets of factual sentences, quotes, and opinions based on their tone labels, thereby varying the emphases placed on different aspects of the same event without altering the underlying collection of facts or altering their wording. We emphasize, however, that the general concept of framing is much broader, encompassing the use of rhetorical strategies, metaphors, or emphasis on particular dimensions of an issue[10]. In equivalency framing, for example, the same information is presented in logically identical but differently worded forms (e.g., "90% success" vs. "10% failure"), thereby also conveying a biased message to the reader but in a somewhat different way than selecting different facts altogether. Other well-known distinctions in the framing literature include episodic versus thematic framing[40] and generic versus issue-specific frames[41].

Results

We first present results for the main pre-registered outcome variable, which measured the positivity or negativity of respondents' feelings towards the article's main subject. Figure 2A displays the average self-reported feelings toward the main subject of the article across three framing conditions (positive, neutral, negative), with individual responses overlaid. To facilitate interpretation of treatment effects, Fig. 2C presents the corresponding average treatment effects, using the neutral condition as the reference group (see "Methods" for details of the regression analysis). As shown, the biased articles significantly influence how people feel about the subject in both positive and negative directions. The positive treatment exhibits an average treatment effect of 3.2 percentage points

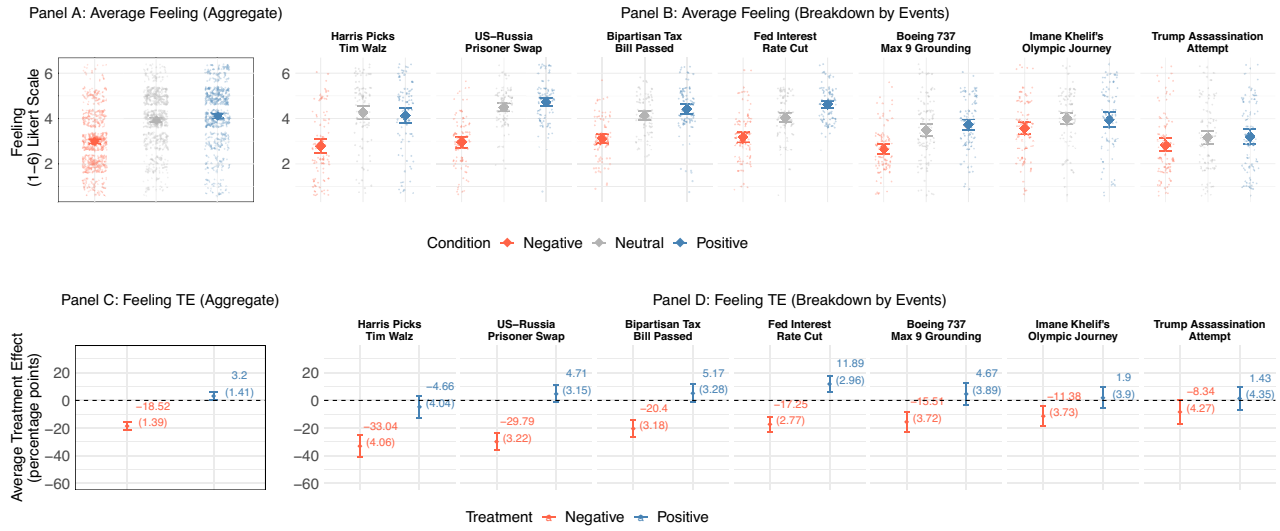


Fig. 2. Framing effects on reported feelings. (**A**) Shows the average reported feelings by condition (positive, neutral, negative) across all events, overlaid with individual response points. (**B**) Presents the same breakdown separately for each event. (**C**) Shows the estimated treatment effects relative to the neutral condition, aggregated across events. (**D**) Displays the treatment effects separately for each event. Error bars indicate 95% confidence intervals. The results highlight that negative framing leads to substantially stronger emotional responses, with consistent patterns across topics.

{95% CI [0.45, 5.96]}, whereas the negative treatment results in an average treatment effect of – 18.5 percentage points {95% CI [– 21.2, – 15.8]}. This effect is not only statistically significant but also represents a relatively large effect size (Cohen's d = -0.73). Notably, there is a significant asymmetrical effect, with the negative impact size being substantially larger than the positive one, as indicated by a χ^2 test of equal effect size ($p < 10^{-9}$).

Figure 2B and D build on Panels A and C by breaking down the results by event. Figure 2B reveals substantial heterogeneity in baseline feelings across events under the neutral condition, reflecting differences in the inherent nature of each topic. Figure 2D further shows that there is considerable heterogeneity across events in terms of the gap between positive and negative treatment effects. This heterogeneity speaks to the importance of sampling multiple events on a diversity of topics, as studies using just a single topic as stimulus cannot be shown to generalize to other equally valid topics. However, Fig. 2D shows that the pattern observed in Panel C–where negative articles elicit stronger responses than positive ones–is broadly consistent across all seven events. In all but one case (Trump Assassination Attempt), the negative treatment produces a statistically significant effect, reinforcing the general asymmetry in impact. In contrast, only one event (*Federal Reserve Rate Cut*) shows a significant positive article effect, with a size of 11.89 percentage points {95% CI [6.1, 17.7]}. Summarizing, Fig. 2 shows how factually accurate but selective reporting of information about an event can influence readers' feelings towards the subject over a broad range of topics.

We note, however, that even if feelings can be manipulated, perceptions about the content of news stories may reflect deeper-held beliefs and therefore be more resistant to manipulation. To explore this possibility, we also asked respondents a series of six event-specific questions that were designed to measure their views on various aspects related to the event they were assigned. Given that each topic is a high-dimensional, multi-faceted phenomenon, and the articles covered a range of points, we crafted questions that capture different facets of each topic. These questions were specifically designed to address various issues referenced in the articles (see SI 1.7 for the full list of questions).

For each event, we conducted a principal component analysis (PCA) to derive a data-driven weighted sum of the questions that effectively captures changes in the opinion space. We chose this approach because our objective was to determine whether individuals generally shift their opinions regarding each event, without focusing on specific aspects. PCA is a robust, data-driven dimensionality reduction technique that allows for the identification of a general direction in the opinion space along which changes can be observed. Moreover, this approach helps mitigate multiple comparison issues by reducing dimensionality and, consequently, the number of statistical tests conducted (see SI 2.3 for details of the PCA analysis and SI 2.4 for results on individual opinion questions).

Figure 3A presents the average of the first principal component scores, separated by framing condition (positive, neutral, negative), with individual response values plotted for reference. And Fig. 3C illustrates the treatment effects on the first principal components, aggregated across the seven different events. The results are highly consistent with those for the feeling outcome: there is no significant effect in the positive direction ($p = 0.2$) but the negative treatment significantly shifts opinion, with an effect size of – 17.94 percentage points {95% CI [– 21.76, – 14.11]}; thus, the asymmetrical effect seen in Fig. 2C persists, with negative treatments exerting a considerably greater impact. Next, Fig. 3B and D break down the results by individual event. Panel B shows the distribution of opinion index scores across conditions for each event, while Panel D presents the

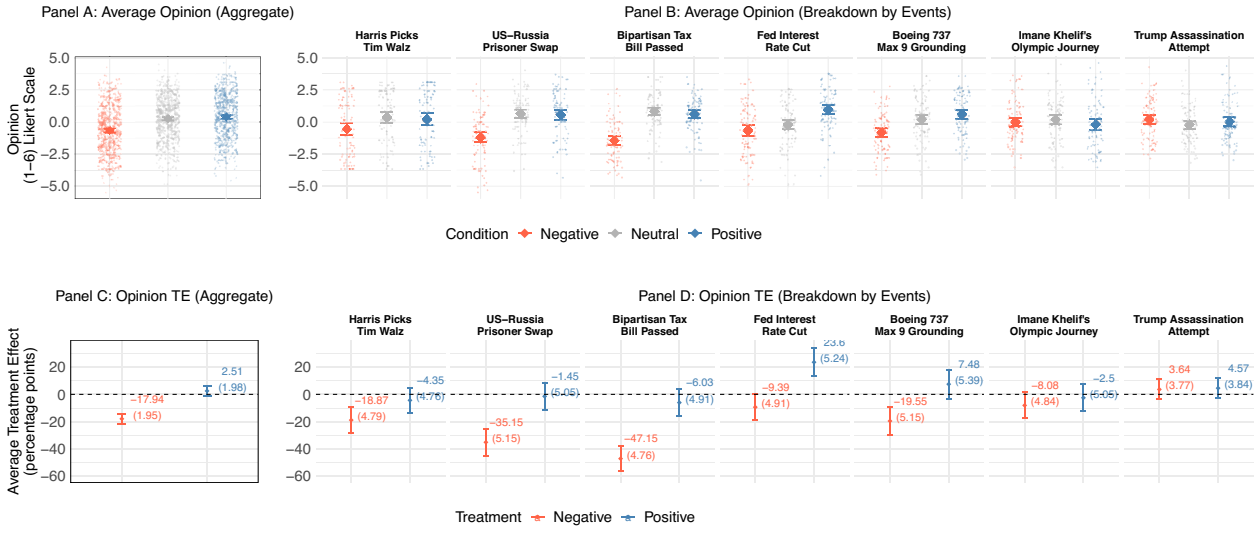


Fig. 3. Framing effects on the first principal component of the opinion outcomes. (**A**) Presents the average opinion scores by condition across all events, with individual-level responses shown. (**B**) Breaks this down by event to show variation in treatment effects. (**C**) Displays the estimated treatment effects on opinions (first principal component) relative to neutral framing, aggregated across all topics. (**D**) Shows these effects for each event separately. Error bars represent 95% confidence intervals. Results reveal that negative framing significantly shifts opinions across diverse policy areas, with generally weaker effects in the positive direction.

corresponding average treatment effects on the first principal component. Once again, there is considerable heterogeneity across events but the overall pattern of negative framing proving more effective than positive framing is the same, albeit with less confidence. Specifically, the point estimates for the negative treatments are negative for six events but statistically significant in only four of them. In contrast, a significant positive effect from the positive treatment is observed in only one event: the *Federal Reserve Interest Rate Cut*. In summary, these results suggest that biased articles not only alter feelings toward the subject but also influence how people perceive the facts associated with these events. For a more detailed analysis of the treatment effect on all individual opinion outcome measures refer to SI 2.4.

To further explore respondents' reactions, we also asked them to evaluate the tone of the article they read. Figure 4A depicts the average tone across different treatment conditions for each event on a six-point Likert scale. These results effectively serve as a manipulation check, indicating that our algorithms successfully generated articles with distinguishable tones, as participants consistently rated the positive and negative articles as significantly more positive or negative than the neutral versions, respectively. Additionally, the figure reveals some imbalance in the difference between the positive and negative conditions compared to the neutral condition, which could contribute to the asymmetry in effects. For instance, for certain events, such as *Harris Picks Tim Walz*, the neutral condition is nearly as positive as the positive condition, likely due to a ceiling effect, as some events are inherently more positive by nature.

To investigate whether the ceiling effect explains the asymmetrical effects between positive and negative treatments, we compare the treatment effects on both feeling and tone. Figure 4B presents the average treatment effects (ATEs) of these two measures side by side for the seven events. The x-axis represents the average treatment effect on tone, while the y-axis represents the effect on feeling, with bars indicating the 95% confidence intervals.

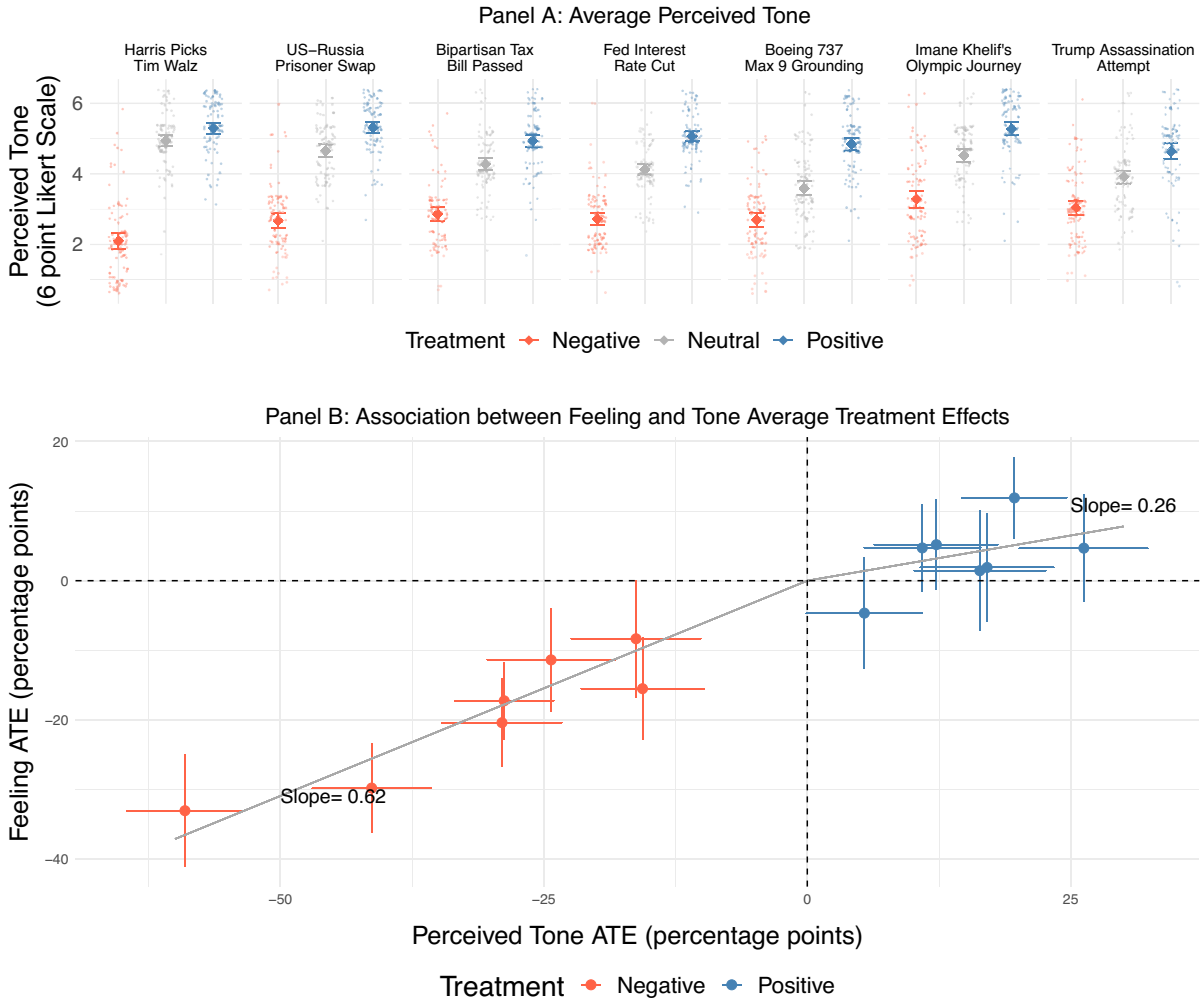


Fig. 4. (**A**) Shows the average perceived tone across treatment conditions on a six-point Likert scale, serving as a manipulation check to confirm the generation of distinct article tones. (**B**) Illustrates the average treatment effects (ATEs) of tone and feeling for seven events, with the x-axis representing tone and the y-axis depicting feeling, alongside 95% confidence intervals. Despite a minor ceiling effect on positive treatments, negative framing consistently exerts a more substantial impact on feelings across similar tone magnitudes, indicating an inherent reader sensitivity to negative news.

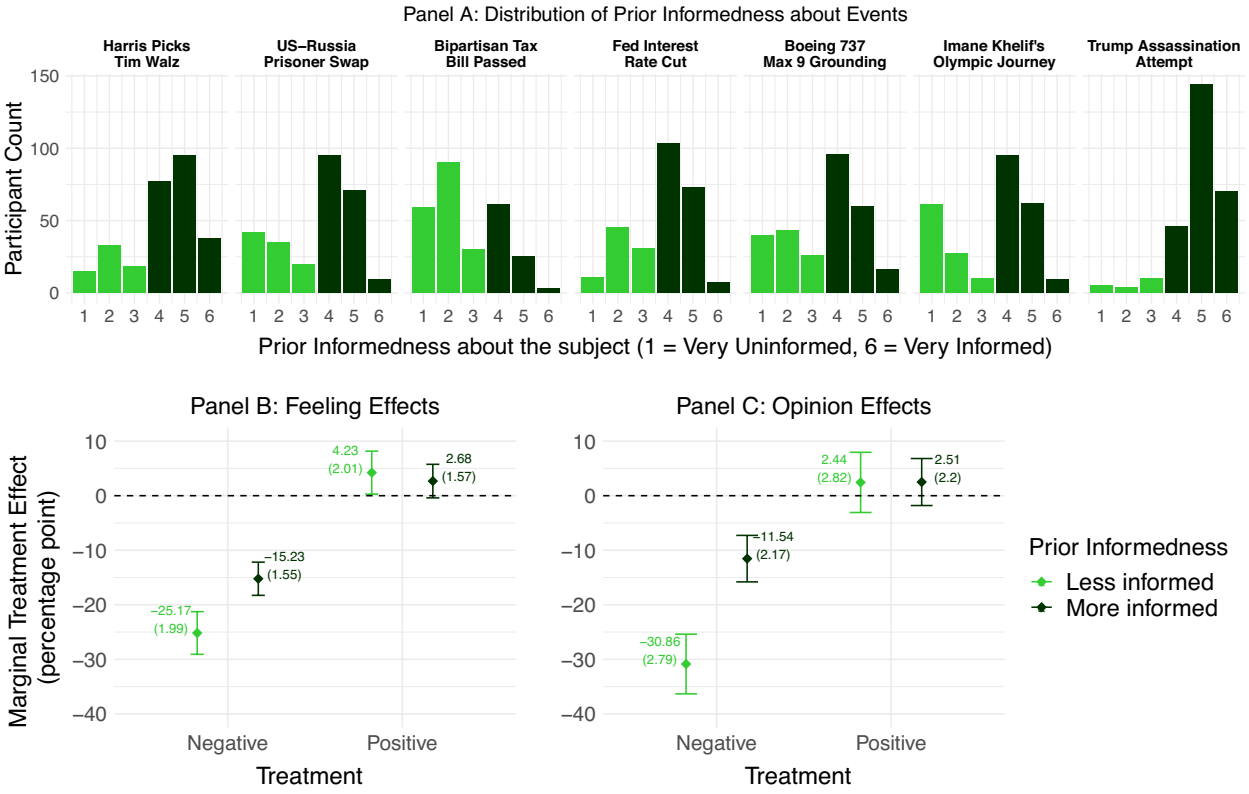


Fig. 5. (**A**) Shows the distribution of participants' prior informedness on a 6-point Likert scale across selected events. (**B**) and (**C**) Illustrate the heterogeneous treatment effects on feelings and opinions, respectively, highlighting the significant moderating role of prior informedness, particularly for negative treatments. Bars represent 95% confidence intervals.

Each event is represented twice, corresponding to the positive and negative treatment conditions, and is identified by the shape of the data points. As observed, there is indeed a limited ceiling effect; the maximum tone effect in the positive direction is 26.2 percentage points for the *Boeing 737 Grounding*, whereas in the negative direction, it reaches – 59.1 percentage points for *Harris Picks Walz*. This asymmetry is partly due to the inherent nature of the selected events, which were generally perceived as somewhat positive even in their neutral versions, as illustrated by the gray points in Fig. 4A. This pattern reflects a combination of our event sampling strategy and the broader negativity bias in news coverage, which results in a relative scarcity of positive statements. See SI Sections 1.2 and 1.5 for further discussion of the sampling process and negativity bias.

Nevertheless, the lower magnitude of perceived positive tone does not explain the whole asymmetrical effect because even when comparing events with similar magnitudes of tone effects, negative framing still yields more substantial impacts on feelings. For instance, the effect on perceived tone for the *Boeing 737 Grounding* negative condition was – 15.6 percentage points, which was smaller than its positive counterpart (26.2). The treatment effect on feeling in the negative condition, however, was – 15.5 percentage points {95% CI [– 22.8, – 8.2]}, while the positive condition did not show a statistically significant effect on feeling, with an effect size of 4.6 percentage points {95% CI [– 2.9, 12.3]}. For a more systematic comparison between changes in tone and subsequent effects on feeling, we fitted two separate regression lines with zero intercepts, assuming zero change in tone should yield zero effects on feelings. The results indicate that, in general, the association between tone and feeling for negative articles is significantly larger than positive ones. Specifically, a 1-point increase in the positive tone of an article leads to an average increase of 0.26-point in how positive people feel about the subject. Conversely, the same change in the negative direction results in a 0.62-point more negative feeling. This suggests that people might be inherently more responsive to negative news, rather the asymmetry being an artifact of the stimuli used. This asymmetry is particularly consequential when considered alongside the well-documented negativity bias in media coverage, which we also see in our dataset (see SI 1.5 for details).

Moderating analyses

As pre-registered, we first examine whether treatment effects were moderated by respondents' prior informedness about each event. Figure 5A shows the distribution of self-reported informedness across seven selected events, measured on a 6-point Likert scale. Participants were grouped into two categories: "Less informed" (scores 1–3) and "More informed" (scores 4–6). We then estimated a heterogeneous effects model by interacting this informedness indicator with the treatment variables.

Figure 5B and C present the effects of framing on feelings and opinions, respectively, across informedness groups. In both cases, prior informedness significantly moderates the effect of negative framing. For feelings, the negative frame has a substantially larger effect on less informed participants, with a 9.94 percentage point difference compared to the more informed group ($p < 0.00001$, χ^2 test). A similar pattern holds for opinions, where the gap reaches 19.31 percentage points ($p < 0.00001$, χ^2 test). The positive frame yields a marginally significant increase in feelings for the less informed group (4.2 percentage points, $p = 0.03$, t-test), but the difference between informedness groups is not statistically significant ($p = 0.47$, χ^2 test).

A potential limitation of this result is that the self-reported informedness was measured after reading the article. The motivation for this design choice was to avoid priming participants prior to reading the article–thus preserving the natural impact of the stimulus and enhancing the credibility of main results. However, a consequence of measuring informedness after treatment exposure is that the former may have been affected by the latter. As conditioning on post-treatment variables can bias moderation, caution is required when offering a causal interpretation of these findings[42]. Nevertheless, we believe the moderation analysis still offers meaningful insights under plausible assumptions. Specifically, if exposure to the article can only increase self-reported informedness (not reduce it), and if this shift does not systematically alter responses to the main outcome measures, then the estimated effects for the low-informed group remain valid. Thus, any upward misclassification of low-informed respondents into the high-informed group likely inflates the estimated effect size for the high-informed group, making the observed difference in treatment effects a conservative estimate of the true moderation effect (see SI 2.2 for further discussion).

We also conducted an exploratory (non–pre-registered) moderation analysis by respondents' partisanship (Independent, Republican, Democrat). While no systematic moderation emerged in aggregate, event-level patterns revealed meaningful heterogeneity for some topics. Relative to Independents, Democrats showed stronger reactions to negative framing for broadly neutral or economic events such as the Tax Bill, Boeing Grounding, and Federal Reserve Rate Cut, but responded more negatively to positive coverage of the Trump Assassination Attempt. Republicans, by contrast, displayed fewer systematic differences (compared with Independents), showing only a marginally stronger negative reaction to the negatively framed U.S.–Russia Prisoner Swap. These asymmetric, event-specific responses likely offset one another in the aggregate, explaining the absence of an overall moderation effect by partisanship. Full results are presented in the SI Section 2.6.

Discussion

Returning to our opening motivation, our study reveals that the selective presentation of truthful information can significantly impact both feelings and perceptions related to real-world events. This effect is not only statistically significant but also represents a relatively large effect size (Cohen's d = -0.73) that exceeds typical effects found in persuasion and framing experiments[43,44] and remains consistent across domains such as politics, the economy, and policy. Importantly, this impact persists even though participants were explicitly informed that the article presents only one perspective and that other perspectives may align with the facts. It is also strikingly asymmetric, with negative framing proving significantly more effective than positive framing, and more pronounced for individuals with limited knowledge about a subject (although we again note that the informedness question was asked after the main intervention and caution is required in making any causal interpretation). Beyond its specific findings, this study also serves as a larger-scale replication of well-established framing effects, extending prior experimental results[45] using a more controlled and transparent stimuli generation process. By leveraging large language models, we demonstrate that classic framing effects and its asymmetrical effect can be reproduced at scale across diverse real-world events, while maintaining consistency in factual content and presentation style.

Our findings also speak to the growing role of automated journalism in contemporary news production. Automated journalism refers to the use of template-based natural language generation (NLG) systems to automatically write news articles from structured data, often with minimal human involvement[46–49]. Initially used for routine tasks such as weather reports and sports recaps[50], automated journalism gained traction in the early 2010s with the emergence of commercial NLG platforms capable of producing increasingly complex and stylistically varied texts. The appeal of automated journalism lies in its ability to generate large volumes of customized or localized stories at a speed and scale unattainable through manual workflows making it particularly attractive to large news agencies and organizations[51]. Our results show that decisions about which data are included and how models are prompted can have significant downstream effects on how events are framed. These choices, often embedded deep within technical or editorial pipelines, may shape public perception not through overt misinformation, but through subtle differences in emphasis and selection. This underscores the need to treat data access and prompt design as consequential editorial decisions, particularly as generative systems are scaled across production environments.

Finally, our study makes a methodological contribution through our use of large language models (LLMs) for information extraction and classification and article generation. A common challenge in prior experimental studies of framing effects, which have relied on manually created stimuli, is the difficulty of manipulating conceptually relevant variables such as tone while keeping all else constant. As a result, treatments can differ on dimensions outside the targeted parameter of interest, undermining the validity of the causal relationships established[52]. Our approach enables precise variation of a single parameter while maintaining constancy across other variables by adjusting the hyperparameters of large language models (LLMs). In this case we varied sentence selection based on tone (positive vs. negative) while preserving variables such as factual accuracy, length, composition, and writing style. However, one could easily adapt our method to vary other variables of interest to study other forms of framing effects. For example, one could vary the veracity of statements through specific prompting, while keeping tone, style, and other factors constant. Or, one could vary the model inputs to change the facts to quotes ratio to generate stimuli that can be used to explore which form of evidence are more

persuasive. Importantly, the speed and consistency of LLMs allows researchers to generate highly controlled stimuli over many contexts, improving the generalizability of the findings as well as their causal validity.

That said, our study is not without limitations. First, as noted earlier, our use of tone to select facts, quotes, and opinions from an underlying corpus of journalistic text is most closely aligned with what is called emphasis framing[38,39], which is only one manifestation of the more general concept of framing. Future work could adapt our framework to examine other types of framing to include the use of rhetorical strategies, metaphors, or emphasis on particular dimensions of an issue[10].

A second key limitation of our approach is the residual variation across stimuli, despite our efforts to control key features. Some of this variation is because of the deliberate design trade-offs made to preserve realism: we instructed the model to assemble articles from original sentences drawn from real news sources, without rewriting them to standardize style or length, and fixed the model temperature to zero to ensure fidelity to these sources. Although we believe this approach enhances ecological validity, it inevitably introduces minor uncontrolled differences–for example, small variations in article length because we fixed the number of sentences rather than total word count. For the experiment stimuli, positive articles are, on average, slightly longer than negative ones (540 words for negative, 601 for neutral, and 610 for positive), though these differences are not statistically significant across events (see SI 1.9 for details). Another consequence of this design is that it conflates the effects of tone and selective fact presentation, since tone guided which sentences were selected for inclusion. We deliberately chose this approach as a first step in order to test the most extreme scenario in which both the tone and informational content of the articles vary simultaneously. However, the same framework could be adapted to hold the informational set constant while varying only linguistic tone, or vice versa–particularly if one is willing to relax the "no rewriting" constraint on sentences. Such extensions would allow future work to experimentally disentangle these dimensions more cleanly.

A third limitation is that we only examined single exposure to biased news whereas, in reality, individuals are exposed to many stories about the same topic or event often over extended intervals of time [53,54]. Although our results demonstrate that even brief exposure can have measurable effects, prolonged exposure may potentially amplify these effects. Alternatively, the effects could be reduced via exposure to other perspectives or via simple attenuation. As the difference between short and long term effects could move in either direction, future research should focus on the impact of exposure to consistently (or inconsistently) biased information over time. Relatedly, our study measures the effects of exposure to biased content on attitudes but does not measure impact on behavioral outcomes such as voting or activism. A more comprehensive approach would involve longitudinal tracking of participants to assess not only changes in their views but also how their behavior may be influenced by prolonged exposure to media bias.

Finally, although we observe substantial heterogeneity in framing effects across topics, the current study design does not allow us to draw firm conclusions about systematic differences between topical domains. For example, we did not explicitly test whether framing is more effective for economic stories than for political ones. Nevertheless, our framework is well suited to scale and support such analyses; however, doing so would require substantially larger samples to ensure sufficient statistical power. We therefore leave this as an important direction for future research.

In conclusion, media organizations are frequently accused of bias, but these accusations also often reflect the biases of the accusers who principally object to news coverage that is inconsistent with their own views [55]. As a result, arguments about bias are inherently subjective, allowing media organizations to claim independence and objectivity based on the accuracy of their reporting [56]. Our approach differs from the usual criticism by creating sets of articles that hold accuracy (as well as topic, composition, and writing style) constant while varying dramatically in tone simply by making different—and on their own defensible—choices about what to include and not include. As a result, we are able to demonstrate both the ease with which bias can be introduced to otherwise typical news coverage as well as its impact on readers without assuming any particular "ground truth" view. By showing how "the sausage gets made"—or, more precisely, one way in which it can get made—our results reinforce prior arguments that a proper understanding of the role of misinformation in society should expand its scope beyond outright falsehoods to include the biased, yet factually correct, reporting practices prevalent in mainstream media [7,35,57]. In particular, the asymmetric impact of negative vs. positive framing is concerning given the well documented negativity bias in news coverage[45,58,59]. Our study builds on prior work that examines the downstream causal effects of negative news on attitudes and perceptions, as well as on consumer demand for negative content[60,61]. Specifically, our contribution is to replicate these earlier findings using a novel, scalable framework that clearly isolates and demonstrates this asymmetry. The combination of higher prevalence and impact of negative news could contribute to persistently negative public opinion topics such as the economy and crime even when objective statistics are strongly positive [62,63]. By clearly demonstrating the ease with which bias can be introduced and its subsequent effectiveness in swaying both feelings and opinions, our approach could be helpful to consumers of news who wish to become more critical readers as well as to producers who wish to guard against bias in the first place.

Methods

Stimuli generation pipeline

We developed a novel pipeline that systematically extracts key information from real news articles and recombines it to generate synthetic, biased frames at scale. This section outlines how we used this LLM-based system to construct controlled stimuli for our experiments and evaluate their effects on public opinion.

Initially, we need to assemble a collection of diverse statements that all discuss the same topic. We utilized the dataset from[35], which compiles homepage top 20 news articles from ten prominent mainstream media outlets across the political spectrum from January to September 2024 – The list includes: Associated Press News, CNN,

The Guardian, The Huffington Post, The New York Times, USA Today, The Wall Street Journal, The Washington Post, Breitbart News, and Fox News.

Considering the fast-evolving nature of news, it is crucial that these statements are published around the same time frame to ensure they form a coherent theme and can be meaningfully combined. To create coherent groups, these articles are clustered into "events," defined as clusters of articles published within a three-day window that exhibit high semantic similarity (see SI 1.1 for further details). These events serve as the foundational basis of our pipeline for generating synthetic biased articles.

The pipeline used for generating these articles is illustrated in Fig. 1, with all prompts utilized at each step detailed in SI 1.3. Our pipeline follows the following steps:

- **Information extraction:**
  - **Step 1**: We use GPT-4o to extract all sentences from the news articles and classify them for *type* (fact, opinion, or quote) and *tone* ("positive," "negative," or "neutral").
  - **Step 2**: We de-duplicate sentences for each (type, tone) pair based on their semantic similarity, as multiple news articles may report identical facts or quotes.
  - **Step 3**: Then we used GPT-4o to rank the sentences based on how informative they are and how well they cover the context of the event. However, we found that informativeness of the sentences was highly correlated with their length.
- **Article generation:**
  - **Step 4**: For each event, we generate three collections of the most informative sentences based on tone, while maintaining a consistent fact/quote/opinion composition. For instance, in the positive condition, this resulted in a set comprising 13 positive facts, 6 positive quotes, and 1 positive opinion. This combination reflects the typical fact/quote/opinion ratio observed across thousands of articles in our dataset. The process is mirrored for both negative and neutral tones.
  - **Step 5**: GPT-4o was tasked with taking these sets of sentences and generating an article in a positive, negative, or neutral tone, ensuring fidelity to the input information.
  - **Step 6**: Finally, we manually reviewed the generated articles to ensure that all input sentences were accurately reflected and that they remained true to the original information. We found the articles to be of high quality, requiring minimal corrections.

For the experiment, we selected seven major events that were extensively covered by the 10 news publishers. Conducted in October 2024, our study predominantly focuses on economic and political events around that time. For a comprehensive list of events, and sampling procedure, refer to SI 1.2. We also conducted a descriptive analysis of the generated stimuli to better understand how each version of an event frames the story and which thematic elements are emphasized (see SI 1.4 for details).

Experiment design

The experiment was conducted on Amazon Mechanical Turk on October 2024 with a curated high-quality sample and involved 2,141 respondents across the seven selected events. The experiment received approval from the University of Pennsylvania's Institutional Review Board (IRB) under Protocol Number 855044 and was preregistered at AsPredicted.org (Pre-registration link: https://aspredicted.org/5mzn-n7dj.pdf). All participants were adults and informed consent was obtained from everyone in advance.

As noted above, we first collected demographic information (age, race, gender, partisanship) and data on respondents' general media engagement. Participants were also presented with instructions indicating that the article they would read presented only one perspective and that other perspectives might be equally consistent with the facts. To ensure they carefully read these instructions, participants completed three comprehension check questions. As pre-registered, respondents who answered two out of the three questions incorrectly were removed from the analysis, resulting in 1,910 participants remaining for the results presented here (for more details, see the SI 2.1).

Participants were then randomly assigned, with equal probability, to one of three experimental conditions (*Positive*, *Neutral*, *Negative*). To prevent potential spillover effects across events, each participant was assigned to only one of the selected events. After reading the article, participants completed a feeling thermometer question which asked them to rate their "feelings toward the main subject of the article", and were also asked about their prior informedness about the subject, and to evaluate the article's tone and informativeness. In addition, they answered a series of six event-specific questions designed to gauge their views on various aspects related to the event. All responses were recorded on a six-point Likert scale. The full list of questions and the synthetic articles can be found in the project's OSF page (OSF link: https://osf.io/9g7sq/?view_only=189a8b9b2f644433bdbfcd4fc5c63ffe).

Regression analysis

Treatment effects are estimated using a linear regression model, with the neutral condition as the control group and the positive and negative conditions as the two treatment arms. The model controls for respondents' demographics and media engagement metrics, and includes event fixed effects for aggregate results (see SI 1.6 for further details). To enhance interpretability, all results are presented in percentage points.

We also report alternative models with standard errors clustered at the event level, and a multi-level model with random effect for events. Furthermore, we perform a robustness check excluding participants who answer

any of the comprehension questions wrong (see SI 2.5 for robustness results). All of the results are robust to these alternative specifications.

Data availability

All survey data, experiment materials, and analysis code required for replicating the results can be found in the project's OSF page (OSF link: https://osf.io/9g7sq/?view_only=189a8b9b2f644433bdbfcd4fc5c63ffe).

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Author contributions

A.T., S.H., and D.W. jointly designed the experiment. A.T. and S.H. conducted the experiment. A.T. analyzed the data and wrote the first draft of the manuscript. All authors contributed to editing and revising the paper. D.W. provided supervision and funding support.

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Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to A.T.

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